

VANSH ARORA

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SUMMARY

AI/ML undergraduate with hands-on experience building real-time computer vision systems and full-stack applications, from model design through deployment. Led operational strategy for a 65-member technical club, coordinating cross-functional teams across multiple flagship events.

EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science & Engineering (Specialisation: AI & ML), CGPA: 8.67

Expected May 2027

Bhopal, MP

Coursework: DSA, OOP, Machine Learning, Deep Learning, Neural Networks, Cloud Computing

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, SQL

Web: React, Next.js, Node.js, Django, Fast APIs, HTML5, CSS3

Databases: MySQL, PostgreSQL (Supabase), Row-Level Security (RLS)

Cloud: AWS (Cloud Architecting, S3) and cloud fundamentals

AI / ML: MediaPipe, OpenCV, scikit-learn, NumPy, Pandas, pytorch, Gemini, Groq, Ollama, Hugging Face, Streamlit, Prompt Engineering, MCP

Tools: Git, GitHub, VS Code, Jupyter Notebook, Figma, Google Stitch

PROJECTS

Smart Fitness Trainer | Python, MediaPipe, OpenCV, Supabase, scikit-learn, pytorch, streamlit

[GitHub](#)

- Built a real-time AI coach that tracks reps and analyses form for squats, bicep curls and Warrior II yoga from a webcam feed (MediaPipe Pose + OpenCV), with asynchronous voice corrections.
- Designed a multi-gate rep-quality engine (tempo, range of motion, extreme angle, posture stability) with penalty overrides: clean reps scored **98.9–99.4/100** vs **17.2–29.1** for bad reps across 10 test scenarios (all passed).
- Scored yoga alignment over 6 joint chains using **atan2** vector trigonometry: **85.5%** for correct Warrior II vs **49.6%** for a defective pose (hold threshold 75%).
- Profiled on CPU at 720p: **16 FPS** inference (P95 83.8 ms), **13.5 FPS** over a 10-min session with **0 crashes** and +6.45 MB memory; on real footage detected 12 curl reps and flagged the 3 fatigued reps at set end.
- Migrated the backend to Supabase Postgres with Google OAuth and Row-Level Security policies; added a per-user upload rate limiter (5/hour) that fails closed on database errors.

Medi-Genie | Next.js, React, TypeScript, Node.js, Express

[GitHub](#)

- Engineered a full-stack healthcare web app offering online doctor consultations, medication reminders and appointment scheduling.
- Built a responsive React/Next.js interface for finding nearby clinics and pathology labs.
- Developed secure REST APIs with Node.js and Express for user authentication and appointment management.

Pahal Interface | Python, Django, MySQL, AWS S3, Razorpay

[GitHub](#)

- Developed a full-stack NGO platform for donation tracking, volunteer coordination and impact analysis.
- Integrated Razorpay payments and AWS S3 storage; implemented role-based access control and a real-time admin dashboard.

Micro-Projects | Python, Jupyter Notebook, NumPy, OpenCV

[GitHub](#)

- Tumor detection in medical scans using a bounding-box localization approach.
- Image denoising pipeline: introduced salt-and-pepper noise, then restored images via filtering techniques.
- Eye detection using an 8x8 patch-grid (64 patches) feature approach.

CERTIFICATIONS

Cloud Computing | NPTEL

Apr 2025

Machine Learning in Python | Coursera

Jan 2026

AWS Certified AI Practitioner | AWS

Aug 2026

Foundations Associate, Agentic AI | Oracle

Aug 2026

ACHIEVEMENTS & LEADERSHIP

- INNOVIT 2026 Hackathon Winner** – built Medi-Genie.
- NPTEL Introduction to IoT, IIT Kharagpur (Apr 2026):** All India Rank 34 (Elite, 98%).
- Bit By Bit Club, VIT Bhopal:** Lead of Operations (Jun 2025 – Apr 2026), coordinated 3 flagship events including a university-wide fest with the highest turnout among 100+ participating clubs; Social Media Lead (Dec 2024 – Apr 2026), led a 6-member content team.

EXTRACURRICULAR ACTIVITIES

- NCC:** Awarded "A" Certificate for leadership, discipline, and adaptability.
- Volunteer, Swachh Bharat Abhiyan:** Participated in community sanitation and awareness drives.
- Project Head, College Technical Exhibitions (x2):** Managed team coordination and event logistics end-to-end.